Sentinel — Plan for Scale

Sizing, deployment and resilience for 80,000 cameras

Plan for Scale · Gujarat Police Innovation Challenge 2026

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Repository: Sentinel · all figures generated from the running system and its benchmark

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Sentinel's statewide design rests on three decisions, each forced by a measurement or a property of the camera estate rather than by preference:

1. **Not every camera needs continuous analytics.** Cameras are tiered by what their position allows, and the registry records it (`anpr_viable`, `analytics_enabled`).
2. **Video stays where it is.** Departments keep recording on their own NVR/VMS. Analytics runs at district edge nodes; only metadata crosses the backbone.
3. **Frames never touch CPU memory at scale.** Our own benchmark shows the CPU, not the GPU, is what runs out first — so production nodes decode and infer on the GPU.

Every sizing and cost number below is generated by `bench/capacity.py` (`python bench/capacity.py`; `--help` lists every input). Each input is labelled **GIVEN** (the problem statement or the PoC's own configuration), **MEASURED** (on the PoC, dated) or **ASSUMPTION**, and all of them are listed in Appendix A. Change an assumption and re-run. Unit prices are placeholders to be replaced by GeM quotes. Anything marked **design** is specified here but not built in the PoC.

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1. What we measured

`sentinel-bench` on the development laptop: GTX 1650 Ti (4 GB), Intel i7-10750H with 6 logical CPUs, 16 GB RAM, Windows 11 + Docker Desktop. Full pipeline per stream: software H.264/H.265 decode → YOLOv11n plate detector (CUDA, FP32) + text-line finder → batched OCR (fast-plate-ocr CCT-XS v2, CPU). Target: 5 analysed frames/s per stream.

Streams	Decoded fps (total)	Analysed fps / stream	Batch latency	GPU util	GPU memory	Host CPU
1 (1080p25)	28	**4.8**	91 ms	26%	171 MB	96%
2 (1080p25)	56	**4.8**	130 ms	33%	171 MB	98%
4 (mixed)	89	3.5	222 ms	26%	253 MB	100%
8 (mixed)	160	1.3	452 ms	21%	493 MB	100%

Source: `bench/results/bench_20260917_174834.json`.

What this tells us

- **The GPU is idle; the CPU is saturated.** GPU utilisation never exceeds a third. Software decode of every frame (inter-frame codecs cannot skip reference frames), colour conversion and image preprocessing consume the host.
- **Confounder, stated plainly:** an antivirus service was using ~41% of the machine at idle during these runs. The absolute numbers are pessimistic; the *shape* — CPU-bound, GPU-idle — is the finding.
- Optimisations already in the worker: grab-every-frame but convert only ~1.5× the analysis rate; batched OCR with capped threads; on-demand browser transcodes (measured 17 Sep: idle media-server CPU from ~130% to ~23% of a core); FP16 auto-disabled on GTX 16xx cards, which return NaN boxes in half precision.
- Hardware decode, measured afterwards on the same laptop (24 Sep, `bench/results/decode.md`, two rounds, plate pipeline without object detection): with D3D11 decode through OpenCV (`HW_DECODE=auto`, now the worker's default) analysed fps per stream rose from 1.93 to 2.88 at 8 streams and from 0.91 to 1.34 at 12 (+49% both), and system CPU at 8 streams fell from 94% to 80%; at 4 streams nothing changed. On this laptop the D3D11 decoder is the Intel UHD integrated GPU's, not NVDEC (Windows GPU-engine counters, 24 Sep; the NVIDIA decoder stayed at 0%). It shows that decode offload relieves the CPU. NVDEC on the node is still unmeasured; on another laptop's GTX 1650, decode only (9 Oct, `bench/results/nvdec_decode.md`), the NVIDIA decoder took about 740 frames a second of 1080p H.264 at its ceiling (about 29 streams at 25 fps) and about 1,500 of 1080p H.265 with the resize on the GPU, and at 16 real-time 1080p streams cut system CPU from 89% to 31% (H.264, decoded and resized to 640 px) and from 75% to 32% (H.265). In today's worker frames still go to the CPU after decode, and a laptop result does not replace the per-node assumption in §6.2.

Consequence for production: GPU-native nodes. Hardware decode (NVDEC via GStreamer `nvcodec` or FFmpeg `cuvid`) keeps frames in GPU memory; preprocessing and inference run on the same device (ONNX Runtime CUDA/TensorRT execution provider); only boxes and crops return to the CPU. The worker's interfaces (`RtspSource`, `PlateDetector`, `PlateOCR`) are the seams where that swap happens without touching the rest of the platform.

Other measurements the model uses (PoC, 24 Sep 2026):

What	Result	Used in the model as
Watchlist match per plate read, 100,000 active entries, one connection	0.79 ms median one character off, 1.14 ms for a plate not on the list (`bench/results/matcher.md`)	`--match-ms 1.14`
Sighting row in PostgreSQL, indexes included	503 B per row in the populated partition, 4,205 rows (631 B if the eleven empty monthly partitions are counted)	`--sighting-row-bytes 600`
Plate crop (JPEG)	6.2 KB average over 4,187 crops	`--crop-kb 8` (conservative)
A sighting as the worker spools it (JSON)	399 B average, the same 4,205 sightings re-serialised in the worker's format	`--spool-row-bytes 400`
Vehicle/person count row	512 B busiest row + ~80 B of index; as spooled JSON, 331 B average and 428 B largest over 368 rows	`--count-row-bytes 600` , `--spool-count-bytes 430`
Store-and-forward	~87 s API outage: 21 sightings spooled, all replayed, alerts raised with the original sighting times	§7.1
Core at statewide size: 80,000 cameras, 3.0 M sightings over 30 days, 100,100 watchlist entries, one API process and PostgreSQL on the laptop	706 sightings/s ingest sustained, each read matched against the watchlist (request p95 718 ms); 30-day movement history 337 ms; map state view as clusters 161 ms (`bench/results/core_scale.md`)	a floor for §11 item 6, not a sizing
The tier B profile (the counts-only worker, keyframes only) on the laptop, 7 Oct: 40 simulator streams at once, no plate worker running, the relay's CPU excluded	0.78 counted frames/s per stream (median) on 2.46 CPU cores, GPU 13%; 50 streams saturate it at 0.34/s (`bench/results/counts_capacity.md`); 7 sandbox cameras 14–34 frames a minute each (`counts_sandbox.md`)	a direction for §6.1; `--node-event-streams` stays 250 (ASSUMPTION)

2. Architecture at scale

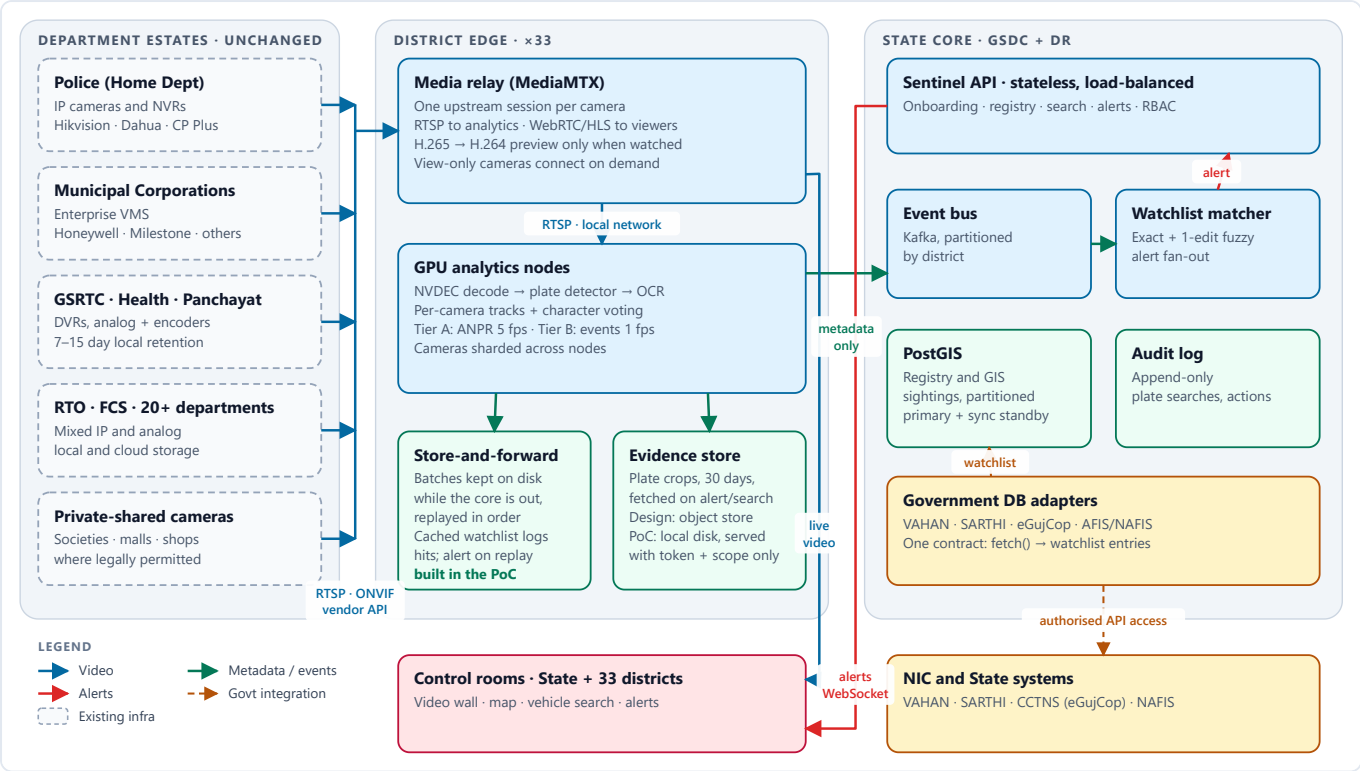


Figure 1. Statewide architecture: department estates untouched, district edges relay and analyse, only metadata reaches the state core.

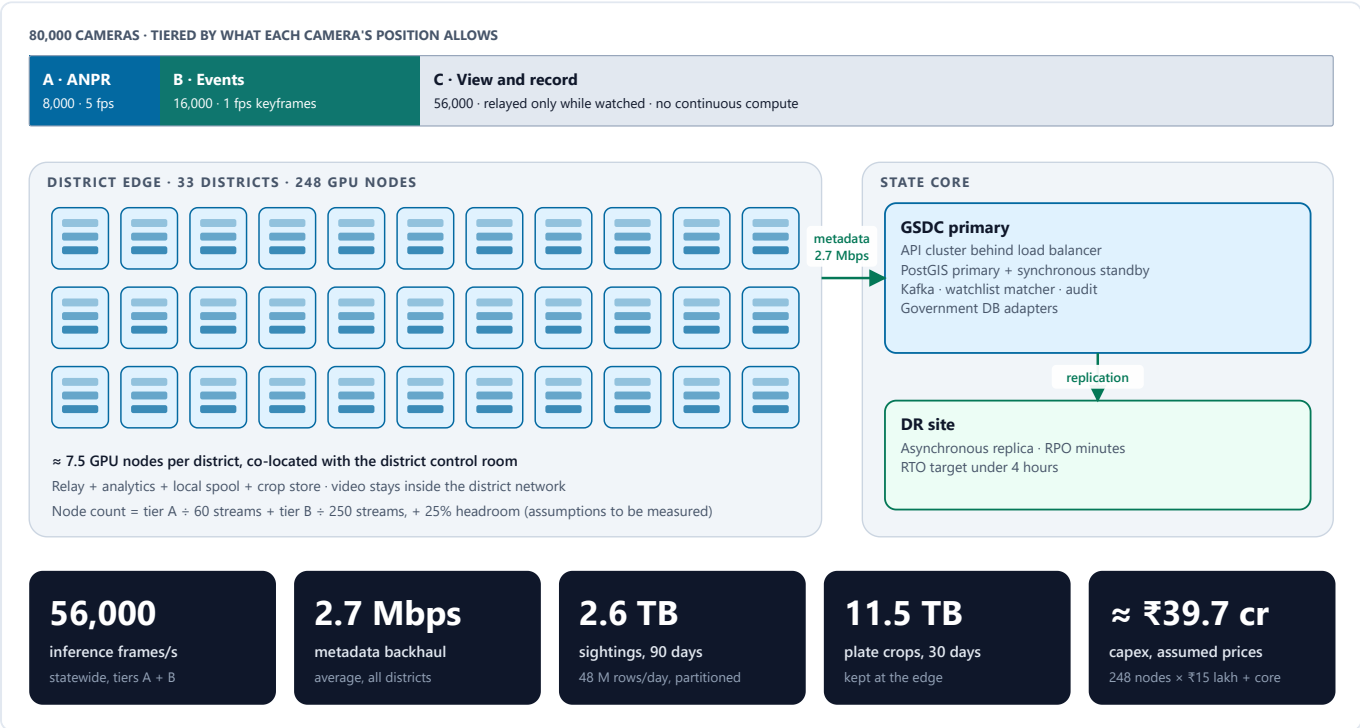


Figure 2. Sizing at a glance. Every figure is generated by bench/capacity.py.

Where compute sits (the portal asks for central, regional and edge requirements):

Layer	Where	What runs there	Sized in
Edge	District sites beside the district control rooms	Media relay, GPU analytics (decode + inference), plate-crop store, store-and-forward spool, cached watchlist for outages	§3.1, §6
Regional	None	Districts connect straight to the state core: the backbone carries metadata (§4.2), too little to justify a regional hop. The range headquarters are a rollout wave (§8), not a tier	—
Central	State Data Centre (SDC) and a DR site	Registry, API, watchlist matching, alerts, search, database, archive, backups, monitoring	§3.2
Control rooms	33 district + 1 state	Browser clients on operator workstations; nothing installed	§3.3

3. Hardware & Software Requirements

3.1 Edge node

The node is a single-GPU server because that is what the capacity model sizes; §6 derives how many.

Component	Specification	Basis
Form factor	2U rack server, redundant power supplies	ASSUMPTION
GPU	1 × data-centre inference GPU with hardware video decode (NVDEC), 24 GB class, e.g. NVIDIA L4	ASSUMPTION: the single-GPU node whose stream capacity the PoC measures first
CPU	1 × 24-core server CPU	ASSUMPTION: frames stay on the GPU; the CPU demuxes RTSP, batches OCR, does I/O
Memory	128 GB	ASSUMPTION
System disks	2 × 960 GB SSD, RAID 1	OS, containers, models
Data disks	2 × 1.92 TB NVMe, RAID 1	86 GB of crops per tier A node for 30 d; spool 1.3 GB for a 7-day outage
Network	2 × 10 GbE (camera side, uplink) + 1 GbE management	ingest at capacity: 240 Mbps (tier A node), 200 Mbps (tier B node)
Load at capacity	tier A: 60 streams, 300 frames/s; tier B: 250 streams, 250 frames/s	ASSUMPTION until measured on the node
Power	~0.5 kW average	ASSUMPTION; feeds the power line in opex
Price	₹15.0 lakh	ASSUMPTION; GeM quote needed

Store-and-forward: a fully loaded node spools ~181 MB a day of sightings and counts; the PoC default cap (`SPOOL_MAX_MB=512`) covers 3.0 days, 1.3 GB covers 7. Production sets `SPOOL_MAX_MB` for the 7-day target. Since 8 Oct each worker process spools under its own folder (`data/spool/<worker id>`, with its watchlist copy and local-alerts file), so a plate worker and a counts-only worker on one node never replay each other's files; the cap is per worker.

Candidate nodes are benchmarked in the pilot with `sentinel-bench`. If another accelerator, or a two-GPU node, wins on cost per stream, the node count and the node price change together. In the PoC, decoded frames are handed to the CPU for preprocessing (§1); keeping them on the GPU from decode to inference is

design.

3.2 Core: State Data Centre and DR site

The core runs no inference: it matches, searches and stores. This bill of materials itemises the ₹2.5 crore core line of §9.1; every unit price is an ASSUMPTION.

Item	SDC	DR	Unit (₹ lakh)	Total (₹ lakh)	Specification
PostGIS database server	2	1	24.0	72.0	2 × 32-core, 512 GB RAM, ~15 TB usable NVMe (RAID 10) for the 4.5 TB hot tier plus indexes, WAL and growth; primary and synchronous standby at SDC, asynchronous replica at DR
Application server	3	2	6.0	30.0	1 × 32-core, 128 GB RAM; API, ingest, watchlist matching, SSO, secrets; stateless behind HAProxy
Event bus broker (Kafka)	3	1	6.0	24.0	1 × 16-core, 64 GB RAM, 4 × 3.84 TB NVMe
Object storage node (Ceph, S3 API)	4	0	8.0	32.0	12 × 16 TB HDD; ~256 TB usable at 3 replicas, for the ≤ 90 TB warm archive and case evidence
Backup repository	0	1	16.0	16.0	pgBackRest target with object lock; the 10.3 TB of base backups and WAL, and audit-log exports
Monitoring and logging server	2	1	6.0	18.0	Prometheus, Grafana, Loki; 1 × 16-core, 64 GB RAM, 8 TB
Firewall (NGFW)	2	2	8.0	32.0	HA pair per site
Core switch, 25 GbE	2	2	4.0	16.0	redundant pair per site
Rack, PDUs, KVM, cabling	3	2	2.0	10.0	per rack; the rack count sets the hosting charge
Core DC + DR				**250.0**	= ₹2.5 crore

The PoC runs all of this as one Docker Compose stack with a single PostgreSQL instance. The HA database, Kafka, Ceph, HAProxy and the monitoring stack are design.

3.3 Control-room workstation

Item	Specification	Basis	Qty	Unit (₹ lakh)
Operator workstation	8-core desktop CPU, 32 GB RAM, discrete GPU with H.264/H.265 hardware decode, 2 × 27-inch displays, 1 GbE, current Chrome or Edge	decodes one 16-tile wall, 17.9 Mbps with 10% of tiles at full resolution; the GPU decodes so the CPU stays free	68	1.5

The control room is a browser application: existing workstations and video-wall controllers are reused wherever they meet this spec, and the 68 new seats in §9.2 are an allowance, not a survey result. Walls show the 640p preview and open one tile at full resolution, because on the PoC laptop a browser could not keep up with many high-bitrate 1080p streams while the worker ran.

3.4 Software stack and licences

Layer	Components	Licence	Status
Edge OS and GPU runtime	Ubuntu Server LTS; NVIDIA driver, CUDA, TensorRT	free (optional paid support); NVIDIA licence, no fee	design (the PoC runs Windows + Docker Desktop, CUDA via PyTorch)
Decode	GStreamer `nvcodec` or FFmpeg `cuvid` on the GPU	LGPL	design (the PoC's plate worker decodes through OpenCV/FFmpeg with a hardware decoder where available, `HW_DECODE=auto`, which on the laptop is the Intel integrated GPU's, and hands frames to the CPU)
Keyframe-only decode (tier B)	PyAV (FFmpeg bindings), the decoder told to skip every non-key frame	BSD-3; the wheel bundles a GPL FFmpeg build, used for decoding only	built 7 Oct (the counts-only worker, §6.1)
Detection and OCR	Ultralytics YOLO (plate and COCO models), fast-plate-ocr, ONNX Runtime, PyTorch, OpenCV	AGPL-3.0, MIT, MIT, BSD-3, Apache-2.0	built
Media relay	MediaMTX, FFmpeg	MIT, LGPL/GPL	built
API and ingest	Python 3.12, FastAPI, Uvicorn, asyncpg, Pydantic, argon2-cffi, PyJWT	PSF, MIT, BSD-3, Apache-2.0, MIT, MIT, MIT	built
Database	PostgreSQL 16, PostGIS, pg_trgm, fuzzystrmatch	PostgreSQL, GPL-2.0	built (single instance)
Database HA and backup	Patroni, pgBackRest	MIT, MIT	design
Cache and live events	Valkey	BSD-3	built
Event bus at scale	Apache Kafka	Apache-2.0	design
Object store	Ceph (S3 API)	LGPL-2.1/3.0	design
Load balancing	HAProxy, keepalived	GPL-2.0	design
Identity and secrets	Keycloak (SSO), OpenBao	Apache-2.0, MPL-2.0	design
Monitoring and logs	Prometheus, node exporter, NVIDIA DCGM exporter, Grafana, Loki	Apache-2.0, Apache-2.0, Apache-2.0, AGPL-3.0, AGPL-3.0	design (MediaMTX already exposes Prometheus metrics)
Front end	React, React Router, Leaflet, react-leaflet, hls.js, Vite	MIT, MIT, BSD-2, Hippocratic-2.1, Apache-2.0, MIT	built
Map tiles	Self-hosted OpenStreetMap tiles	ODbL (data)	design (the PoC uses the public tile server)

The cost model carries **no software licence fees** (§9.3). Two licences carry obligations:

- **Ultralytics YOLO (AGPL-3.0).** A government deployment publishes its changes under a compatible licence, or buys a commercial Ultralytics licence (not priced here), or retrains both detectors under a permissive framework. Both detectors run on Ultralytics YOLO, licensed AGPL-3.0. A licence-clean replacement is planned for the pilot, not the finale: YOLOX detectors (Apache-2.0) run by ONNX Runtime (MIT), with the plate model retrained only on images whose licences allow commercial use. Nothing has been trained. It replaces the current models only if it passes adoption gates we wrote down before training (`bench/detector_gates.md`): no loss of plate capture or exact reads on our own feeds, no loss of plate yield on the sandbox cameras, and no slower analysis. Until then the AGPL models are what runs, at the finale too, and every number in this pack was measured with them. We never quote the plate model's published scores: its own model card says they are inflated by an overlap between its training and test images. Every accuracy figure here is our own measurement. The plan covers the detectors only; the OCR model's training data is not documented by its author. Grafana and Loki are also AGPL-3.0, but they run unmodified as separate services, so their licence does not reach Sentinel's code.
- **react-leaflet (Hippocratic License 2.1)**, the map bindings in the PoC's front end. It is not an OSI-approved open-source licence: it requires use consistent with the UN human-rights principles and settles disputes by public arbitration. A police deployment has it cleared by legal review, or replaces it with direct calls to Leaflet (BSD-2).
- **PyAV (BSD-3) with a bundled GPL FFmpeg.** The counts-only worker (7 Oct) decodes through the PyAV wheel, whose FFmpeg is a GPL build, used for decoding only. A deployment builds PyAV against its own LGPL FFmpeg, or accepts the GPL terms as it already does for the FFmpeg in the relay image; on the edge node the GPU decode of row 2 replaces it in any case.

4. Network & Bandwidth Planning

4.1 Camera → edge

Leg	Cameras	Per camera	Average district	Statewide
Tier A: main stream, continuous	8,000	4 Mbps	970 Mbps	32.0 Gbps
Tier B: sub-stream, continuous	16,000	0.8 Mbps	388 Mbps	12.8 Gbps
Tier C: only while watched	56,000	0 when idle	≤ 17.9 Mbps (one 16-tile wall)	≤ 609 Mbps (every wall)
Continuous total	24,000		**1.36 Gbps**	**44.8 Gbps**

Video stays inside the district network for analytics, but this leg is not small: the analysed cameras put about 1.36 Gbps into an average district's edge. Averages hide the skew, since cameras concentrate in the cities, so each district's network is sized from its registry count during onboarding. Three consequences:

- The edge rack needs 10 GbE aggregation from the camera VLANs, priced as district network works in §9.2 and confirmed by the site survey.
- A camera behind a thin link (a remote checkpost on a cellular link, say) cannot be analysed at the district edge at 4 Mbps. For those sites the design option is a small edge box beside the camera or NVR, not priced here.
- Tier B pulls the camera's sub-stream. The object detector already resizes every frame to 640 px (`object_imgsz`), so a 640p stream gives it about the same pixels at a fifth of the bitrate. Tier A needs the main stream: the plate is read from the full-resolution crop.

4.2 Edge → core and viewing

Flow	Rate	Note
Edge → core: metadata	2.7 Mbps	average; 556 sightings/s
Edge → core: plate crops	36 Mbps	or keep crops at edge, pull on alert/search (→ ~0)
Edge → core: vehicle/person counts	1.9 Mbps	400 rows/s: one per analysed camera per 60 s
Control-room viewing	609 Mbps	peak; 544 tiles, 10% at full resolution; district rooms sit beside their edge, so only the state room's wall (17.9 Mbps) crosses the backbone
SDC → DR: database replication	9.2 Mbps	average WAL, 99 GB a day at 2× the stored bytes
Backhaul per district, peak	19.1 Mbps	its share of metadata, crops and counts, plus the state room watching 16 of its cameras; links: 2 × 50 Mbps (5.2× headroom)

Low-bandwidth operation: the backbone carries metadata, not video. Browser preview uses a 640p H.264 profile (the camera's second stream where available, an on-demand transcode where not); full resolution is fetched only for the tile an operator opens. A view-only camera costs no bandwidth until someone watches it. When the backhaul drops, the edge keeps working: sightings spool to disk and replay in order when the link returns. In the meantime watchlist hits against the edge's cached copy of the watchlist are recorded at the edge (the worker's log, plate masked, and `data/spool/<worker id>/local-alerts.jsonl`) and reach the control room on replay, with the original sighting time (§7.1).

4.3 The alternative: carry and record every stream centrally (Model 4)

At the same per-camera bitrates used above:

	Model 4 (central VMS)	This design
Backbone, sustained	320 Gbps (80,000 × 4 Mbps)	2.7 Mbps metadata; 40.1 Mbps with crops and counts
Recording, 15 days	51.8 PB; 103.7 PB with a DR copy	none: video stays on department NVRs; 16.0 TB of sightings, counts and crops
Floor: record only the 0.8 Mbps sub-stream	64 Gbps; 10.4 PB (20.7 PB with DR)	

The 15 days is an ASSUMPTION: the upper end of what departments keep today. Continuous recording at a constant bitrate is also an assumption: motion-triggered or variable-bitrate recording would shrink the Model 4 storage figures, and even the sub-stream floor needs 10.4 PB.

5. Storage & Retention Strategy

Data	Per day	Retention	Total
Video	—	stays on department NVR/VMS (7–15+ days)	not duplicated
Sightings (Postgres, partitioned)	48 M rows / 29 GB	90 d hot	2.6 TB
Plate crops (object store at edge)	384 GB	30 d	11.5 TB
Vehicle/person counts (Postgres, partitioned)	35 M rows / 21 GB	90 d hot	1.9 TB
Warm archive (Parquet on the core object store)	—	5 years	≤ 90 TB (upper bound: database bytes, before compression)
Backups (base backups + WAL, at DR)	99 GB WAL	2 weekly fulls, 14 d point-in-time	10.3 TB
Edge spool (store-and-forward)	~181 MB per node	until replayed	1.3 GB per node for 7 d
Alert evidence	small	per case / legal hold	cold archive
Audit log	small	legal retention	WORM export (object lock)

Vehicle and person counts are one row per analysed camera (tiers A and B) per minute; the row size is measured on the PoC (512 bytes for the busiest camera, plus indexes). The window length is the lever: five-minute windows (`--count-window-s 300`) cut the table to 7 M rows and 4 GB a day, at the cost of coarser peaks.

Hot, warm and cold:

- **Hot** (PostgreSQL, 90 days): sightings and counts are range-partitioned by time (monthly in the PoC, daily at statewide volume), so retention is `DROP PARTITION`, not a delete job. The PoC creates the partitions; the scheduled job that drops expired ones is design.
- **Warm** (object store, 5 years): older partitions export to Parquet for yearly investigation queries (design).
- **Cold**: evidence linked to an FIR is copied to a cold archive under legal hold; the audit log is exported daily under object lock (§7.2). Both design.
- **Video** is never copied: it stays on department recorders under their own retention, and the registry records each camera's retention days so an investigator knows how long footage exists.

Retention periods are the Home Department's decision, not the software's; the values here are the model's ASSUMPTIONS.

6. AI Processing Capacity

6.1 Analytics tiers

Tier	Share	Cameras	Analysis	Continuous compute
A — ANPR	10%	8,000	5 fps, full decode	yes
B — event analytics	20%	16,000	1 fps, keyframe-only decode	yes (light)
C — view & record	70%	56,000	none; relayed only while watched	no

Inference load: **56,000 frames/s** statewide (tier A 40,000, tier B 16,000).

Tier shares are **ASSUMPTIONS**. The registry replaces them with fact as departments onboard. Tier B decodes keyframes only (`skip_frame=nokey`): at typical 1–2 s GOPs that gives 0.5–1 analysed frames/s at a fraction of the decode cost, and the model sizes tier B at the upper end, 1 fps.

This profile was built on 7 Oct as the counts-only worker (`sentinel_analytics/counts.py`; HLD 5.4) and measured on the development laptop: 40 simulator streams at once at 0.78 counted frames/s per stream (median) on 2.46 CPU cores and 13% of the GPU, with no plate worker running and the relay's CPU excluded; 50 streams saturate it at 0.34/s (`bench/results/counts_capacity.md`). The same evening it counted 7 of the organisers' sandbox cameras at 14–34 frames a minute each while the sandbox dropped every session about every 45 s (`counts_sandbox.md`). Keyframes 1–2 s apart cap the rate at 0.5–1 frame/s, as the model assumes; a camera whose keyframes are more than 4 s apart (three sandbox cameras keyframe every 10–15 s) is decoded in full so its zones do not miss short presences, and then costs tier A's decode, which the model does not price. Per-frame decode measured on the laptop with 14 simulator streams: 0.47 CPU cores keyframes-only against 2.57 decoding every frame (`bench/results/keyframe_decode.md`, 8 Oct; the 7 Oct probe gave 0.52 and 2.53). The 250 streams per node of §6.2 remains an ASSUMPTION: the laptop's CPU, not its GPU, ran out at 50 streams, NVDEC was not used, and a laptop figure is a direction. Unmeasured: the accuracy of the profile's counts (the same per-frame COCO detections, at about one frame a second), and the load of both profiles on one machine. Before 7 Oct a tier B camera ran the same worker as tier A, plate path included; a plate worker given a wide camera still does, and yields few plate reads, not none (the wide stock camera read 21.7% of the plates on screen for a second or more; HLD 7.5).

6.2 Edge compute and GPUs

Item	Value	Basis
Tier A nodes	134	8,000 / 60 streams per node (ASSUMPTION)
Tier B nodes	64	16,000 / 250 streams per node (ASSUMPTION; the laptop counted 40 at once, §6.1)
Headroom	25%	N+1 failover and growth
Edge GPU nodes	**248**	≈ 7.5 per district, co-located with district control rooms
GPUs at the core	0	the core matches, searches and stores; it runs no inference
Watchlist matching at the core	0.63 busy connections	556 reads/s × 1.14 ms median (MEASURED at 100,000 entries)

The GPU requirement is therefore 248 single-GPU nodes (§3.1), one GPU each. Per-node stream capacity is the most important assumption in this document and the first thing the PoC measures: run `sentinel-bench --streams 8,16,32,64` on the candidate GPU-native node and replace `--node-anpr-streams`, and `bench/counts_capacity.py` for `--node-event-streams`. Watchlist matching, measured at 100,000 entries, keeps less than one database connection busy at the statewide average read rate. On one laptop, a database of 80,000 cameras and 3.0 million sightings took 706 sightings/s of ingest, each read matched against 100,100 watchlist entries, against the 556/s average (`bench/results/core_scale.md`, 24 Sep): a floor, not a sizing (§11).

Load balancing and horizontal scaling. Workers shard cameras across nodes by index (`WORKER_INDEX` / `WORKER_COUNT`, built), so capacity is added by adding nodes; a worker on another machine than the relay sets `RELAY_HOST` (built 7 Oct), since the registry stores relay addresses as `localhost`. The counts-only worker takes every analytics camera the plate workers do not (`COUNTS_EXCLUDE`), up to `COUNTS_MAX_STREAMS`. Today the shard

is fixed when a worker starts; moving a failed node's cameras to its peers automatically is design (§7.1). The API is stateless and scales behind HAProxy (design); PostgreSQL adds read replicas for search and moves to daily partitions at statewide volume (design).

6.3 Sensitivity: the two levers

Case	Edge nodes	Capex (₹ crore)	5-year TCO (₹ crore)	₹ per camera per month
As planned	248	39.7	100.7	210
30 tier A streams per node	414	64.6	141.7	295
45 tier A streams per node	303	48.0	114.2	238
90 tier A streams per node	192	31.3	86.8	181
Tier A share 5%	164	27.1	79.9	166
Tier A share 15%	330	52.0	120.9	252

Each 1% of cameras moved from tier C into tier A adds 16 nodes (₹2.4 crore capex, ₹4.0 crore over 5 years). Both levers are decided by facts not yet in hand: the node measurement and the tier shares the registry reveals.

7. Disaster Recovery Strategy

7.1 High availability and failover

Component	Failure	Behaviour	Status
District backhaul or API	link or API down	Workers spool batches to disk, keep matching plates against a cached copy of the active vehicle watchlist and record local alerts; on reconnect they replay oldest first with the original frame times, and the control-room alert is raised on replay	**Tested** 24 Sep: ~87 s API outage, 21 sightings spooled and all replayed, alerts raised with the original sighting times
Media relay	restart	API re-registers every relay within one health interval	**Tested**: 2/2 relays restored after a forced MediaMTX restart
Camera / NVR	offline	Health monitor flags it within one interval (20 s); uptime and gap reports record it	Built
Camera stream	keeps dropping the session	The worker reconnects with backoff from 2 s to 30 s, and starts again from 2 s only after 60 s of healthy streaming, so a source that drops every session a few seconds in is not retried every 2–5 s (on 25–26 Sep the sandbox dropped our sessions three times and twice refused our login)	Built 26 Sep; 2 unit tests
Edge analytics node	node down	Its cameras move to peers within the 25% headroom	Design: the shard is fixed at worker start today, so peers take the cameras when restarted with a new `WORKER_COUNT`
A worker on another host	the relay and the workers on different machines	`RELAY_HOST` points a worker at the relay machine; `CORS_ORIGIN_REGEX` at the API and `WEBRTC_HOST` at the relay let a control room on another machine sign in and play video (`scripts/lan.ps1` writes both and restarts the relay and API)	Built 7 Oct; the finale runs the plate worker on one laptop and the counts-only worker on a second
The single core host (PoC)	the machine running the database, API and relay fails	Hot spare by hand: `scripts/db_backup.ps1` dumps the database and copies the crops to a USB stick every 10 minutes; `scripts/db_restore.ps1` restores them on the spare and starts its API, whose relay rebuilds every camera path from `camera_sources` within about one health interval (20 s); workers are repointed with `API_URL` and `RELAY_HOST`. Up to 10 minutes of sightings are lost; the standby below is the statewide answer	Built 8 Oct (scripts); the drill is in the finale runbook
API	instance down	Stateless behind HAProxy	Design (the PoC runs one instance)
PostGIS primary	host down	Synchronous standby promoted (Patroni); no committed data lost within the SDC	Design (the PoC runs one instance)
State Data Centre	site loss	Asynchronous replica at the DR site promoted; RPO minutes, RTO under 4 h	Design; RPO and RTO are targets, measured by the restore drill below

7.2 Backup and restore

Item	How	Status
Continuous WAL archiving, point-in-time recovery	pgBackRest ships WAL from the primary to the DR repository as it is written; any moment in the last 14 days can be restored	Design
Base backups	Weekly full to the DR repository, encrypted; two fulls kept, 10.3 TB with 14 days of WAL (§5)	Design
Warm archive	Parquet on the SDC object store, three replicas. There is no copy at DR in this bill of materials, so a site loss would lose sightings and counts older than the 90-day hot tier until one is added (≤ 90 TB after five years, §5)	Design; DR copy not priced
Audit log, WORM copy	Nightly export of the day's `audit_log` rows to the object store under object lock (compliance mode), each day's file hashed together with the previous day's hash, so any later edit is detectable	Design. Built: database rules make UPDATE and DELETE on `audit_log` no-ops
Restore drill	Quarterly: restore the latest base backup and WAL to a scratch host, run the 34-check end-to-end smoke test against it, and record the time taken as the measured RTO	Design
Edge nodes	Stateless apart from crops and the spool; rebuilt from a golden image. Crops tied to an alert or case are copied to the core	Design
Registry and relay configuration	Held in the database; relays are rebuilt from it	Built: relays re-register from the registry after a media-server restart

7.3 Monitoring, logging and health checks

Signal	How	Status
Camera health	Every 20 s: relay readiness and byte flow from MediaMTX, or a TCP probe; results in `camera_health_checks`, the Health page, uptime and gap reports	Built
Plate-reading quality	Every stored plate capture graded good, marginal or poor from its crop (size against published guidance, blur, light, contrast, layout, reader confidence); per camera a tier (reads plates, marginal, counts only) with camera, model and configuration advice, on the Health page (HLD 5.3)	Built 26 Sep
Relay self-healing	Missing relays re-registered every interval	Built; tested 2/2
Worker liveness	Heartbeat every 15 s with achieved fps per camera → `cameras.last_seen_at` and a live event	Built
API liveness	`GET /healthz` for the load balancer	Built
Store-and-forward	Spooling, replay, dropped batches and local alerts in the worker log; local alerts also in `data/spool/<worker id>/local-alerts.jsonl`	Built (log only)
Media-server metrics	MediaMTX Prometheus endpoint, bound to the host	Built; not yet scraped
Metrics and alerting	Prometheus scrapes API, workers, MediaMTX, node exporter and the NVIDIA DCGM exporter; Grafana dashboards; alert rules for node down, spool growing, GPU saturated, backhaul down, replication lag	Design
Logs	Structured logs shipped to Loki at the core; each edge buffers locally while its link is down	Design (the PoC logs to the console and files; since 8 Oct the API's access log is off and the worker's sighting and local-alert lines mask the plate, HLD 11.2)
Security events	Audit log and firewall logs exported to a SIEM	Design (roadmap)

7.4 Security controls

Implemented in the PoC:

- Role-based access (admin, supervisor, operator, department nodal officer, auditor) with department and district scoping enforced in queries, not in the UI
- Append-only audit log of every login, plate search, movement query, onboarding and alert action; database rules make UPDATE and DELETE no-ops
- Camera credentials never reach browsers — everything reads the relay; stream URLs are credential-masked in audit records and, since 8 Oct, in the registry listing and CSV export every reader sees; the onboarding form hides the pasted address while a screen is projected, and `scripts/onboard_urls.py` takes a list of addresses through the relay so the credentials stay in `camera_sources`
- Internal ingest authenticated separately from user tokens, the key compared in constant time; a batch is capped at 5,000 sightings, and an event time without a timezone or far from now is refused

- Sign-in: Argon2id verified off the event loop, more than 30 attempts a minute from one address refused before the password is checked (no lockout or MFA yet); an error logged at every start while a secret is still an example value, with `scripts/rotate_secrets.py` and `scripts/set_passwords.py` to replace them
- Registry input validated on every write path (API, CSV import, catalogue sync): a camera code is letters, digits, `-` and `_`, a stream field is a stream address; the worker opens rtsp and http addresses only; enumerated query parameters answer 422 to a bad value
- Audit also covers listings of a camera's recent sightings, watchlist checks of one plate and registry exports (8 Oct); a user scoped to a district alone is confined to it
- The API's access log is off and the worker's sighting lines mask the plate (HLD 11.2)
- Evidence crops served only with a valid token, the right permission and camera scope; there is no open static folder
- PostgreSQL, Valkey, the media server's control API and metrics, and the simulator's control port bound to the host. Pushing a stream into the relay needs the publisher credential; an anonymous publish from the LAN is refused (checked 24 Sep), and a path already being published cannot be taken over (`overridePublisher: no`). Viewing a stream is still open on the local network: the relay authorisation hook below closes that
- 23 negative security checks (`scripts/security_checks.py`) and 309 unit tests cover the above (HLD 11.2)
- Face recognition deliberately absent (see privacy-dpdp.md)

Production additions (design):

- Camera VLANs with no internet egress; edge nodes are the only bridge
- mTLS between edge and core, TLS for every browser session; secrets in a vault (`credentials_ref` already modelled); encrypted disks on edge nodes, which hold crops and the spool outside the data centre; encrypted backups
- Media relay authorisation delegated to the Sentinel API per path (MediaMTX HTTP auth hook), so RBAC governs who can watch which camera and nothing but the relay's own sources can publish
- CERT-In incident reporting integration; STQC / third-party audit before go-live (priced in §9.2)

8. Statewide Rollout Plan

Wave	Scope	What it proves	Exit criteria
0 — Pilot district (0–3 months)	One district, all departments, one GPU-native edge node	Real accuracy per camera class; real VAHAN/eGujCop access; per-node capacity measured	Live test case passes on that district's estate; node capacity replaces the assumption in §6.2
1 — Range headquarters (3–6 months)	Ahmedabad, Gandhinagar, Surat, Rajkot, Vadodara	Density and mixed-vendor integration in the largest estates	Registry complete for those districts; control rooms using it daily
2 — Highway and border corridors (6–9 months)	Checkposts, toll approaches, border districts	Interception value: the corridors where a tracked vehicle actually travels	Cross-district movement histories in routine use
3 — Remaining districts (9–18 months)	District by district, in registry order	Steady-state onboarding at wave cadence	80,000-camera registry; tier assignments from real geometry

Each wave follows the same sequence: collect department information (§10), survey the district network (§4.1), register cameras, connect the edge node, tier the cameras, enable analytics on tier A, train the control room, then review the gap analysis before the next wave.

9. Implementation and Operational Costs

All figures are ASSUMPTIONS from `bench/capacity.py`, to be replaced by GeM quotes and the pilot's measurements. They are here so the cost structure can be checked and re-run, not as a quotation.

9.1 Hardware capital cost

Item	Qty	Unit (₹ lakh)	Total (₹ crore)
Edge GPU nodes	248	15.0	37.2
Core DC + DR	1	250.0	2.5
Total			**39.7**

The core line is itemised in §3.2. Two levers dominate cost and both are policy or measurement, not technology: the tier A share and per-node stream density (§6.3).

9.2 One-time implementation

Item	Qty	Unit (₹ lakh)	Total (₹ crore)	Basis
Department integration	26	10.0	2.60	adapters, credentials, firewall rules, registry import, per department
Pilot district (wave 0)	1	30.0	0.30	field survey, ground-truth labelling for accuracy per camera class, travel
Production hardening	3	18.0	0.54	FTE-years: GPU-native decode, HA database, event bus, SSO, relay authorisation
Edge installation and commissioning	248	0.5	1.24	per node
District network works	33	10.0	3.30	10 GbE aggregation and fibre from camera VLANs to the edge rack
Control-room workstations	68	1.5	1.02	2 per room where existing ones fall short of the spec
Training	34	2.0	0.68	per control room, incl. train-the-trainer
Security audit before go-live	1	25.0	0.25	STQC / CERT-In empanelled auditor
Total			**9.93**	

Private cameras (societies, malls) join through the same onboarding; their integration is not priced separately.

9.3 Annual operating cost

Item	₹ crore a year	Basis
Edge power	1.39	248 nodes × 0.5 kW × PUE 1.6 × 8,760 h at ₹8/kWh = 1.74 GWh
State Data Centre hosting	0.30	5 racks (SDC and DR), power included
Platform team	1.80	10 FTE; control-room operators are existing police staff, not counted
Field engineers	0.88	11, one per 3 districts
District backhaul	1.98	2 × 50 Mbps per district, two carriers
Core links	0.60	SDC uplink, DR uplink, SDC-DR replication
GPU refresh reserve	1.98	248 GPUs × ₹4 lakh every 5 years
Software licences	0.00	open-source stack; optional vendor support not included
Years 1–3	**8.93**	hardware under warranty
AMC from year 4	3.18	8% of ₹39.7 crore capex
Years 4–5	**12.11**	

The model is deliberately conservative in three places: operating cost runs at full scale from year 1, although the rollout takes 18 months (§8); the GPU refresh reserve is charged every year, although no card falls due for replacement within the five years; and SDC hosting is charged, although a Home Department tenant may pay nothing. District backhaul may also ride on the existing state network rather than new links. Not included: security audits after go-live, vendor support subscriptions, maintenance of the workstations, a DR copy of the warm archive (§7.2), and any charge for VAHAN or eGujCop access.

9.4 Five-year total cost of ownership

Component	₹ crore
Hardware capex	39.7
One-time implementation	9.9
Operating cost, years 1–5	51.0
5-year TCO	**100.7**
Per camera per month	₹210 (all 80,000 cameras)
Per analysed camera per month	₹699 (24,000 tier A + B cameras)

9.5 What is *not* bought

The largest saving is the spend this architecture avoids. A Model 4 central VMS would re-ingest and re-record all 80,000 streams centrally (§4.3):

Avoided by integrating instead of replacing	Scale of the avoided cost
Central recording storage	51.8 PB of primary storage for 15 days (103.7 PB with its DR copy), duplicating what departments already record locally
Statewide ingest bandwidth	320 Gbps sustained to carry every main stream to a central site (64 Gbps even for sub-streams only), against ~2.7 Mbps of metadata in this design
Camera and recorder replacement	Existing cameras, NVRs, DVRs and VMS licences are reused; no department is asked to re-procure
Per-camera VMS licensing	No new per-channel licence: the platform reads one stream per camera over open protocols
Migration downtime	Departments keep operating their own systems throughout; there is no cut-over

The model does not price the Model 4 alternative; the quantities above are what it would have to buy.

9.6 Benefits

Benefit	How it is realised	How to measure it
A vehicle can be followed across departments	One searchable movement history from every ANPR-viable camera, whoever owns it	Time to produce a movement history for a registration number: today's baseline (footage requested department by department) to be measured in the pilot; on the PoC, 337 ms for a 30-day history over 3.0 million sightings (`bench/results/core_scale.md`)
Stolen and wanted vehicles detected while still moving	Watchlists from VAHAN and eGujCop (mock adapters in the PoC) matched against every plate read as it happens	Alerts raised per month, and time from plate read to control-room alert. On the PoC, first valid read at the worker to the alert row in the API: median 1.06 s, p95 1.94 s over 31 alerts (24 Sep, 4 cameras, worker and API on one laptop, same clock; `bench/results/latency.md`). That includes the confirmation hold (a new plate track waits until three reads have arrived or a window of one second, longer on slowly analysed cameras, closes) but not camera-to-edge transport or the browser; to be measured end to end on live feeds
The state knows what it owns	One registry with health, location, ownership and retention	Share of the estate registered and online; gap-analysis findings closed per quarter
Cheaper, faster investigation	Evidence crop, timestamp and location per sighting, exportable for the case file	Officer-hours per investigation involving CCTV
Procurement leverage	Vendor-neutral by construction; departments can buy any ONVIF/RTSP camera	Number of distinct vendors integrated without custom work
Accountability	Append-only audit of plate searches, movement queries and every operator action	Audit review findings; percentage of searches carrying a stated purpose

10. Department-wise information requirements

Onboarding is only as fast as the information each department supplies. This is the complete ask. Its per-camera rows (inventory, stream access, make, model and codec, retention and storage, which cameras can read plates, ownership) are columns of the CSV template in the platform (Registry → Template); the department-level rows (nodal officer, credentials policy, network path, recorder or VMS details) and the installation and firmware details are collected on a department sheet.

What we need	Why	Blocking?
Nodal officer: name, phone, email	One accountable contact per department for access and incidents	Yes
Camera inventory: code, name, latitude, longitude, address	The registry and the map; a camera without a location cannot be searched geographically	Yes
Stream access: RTSP/ONVIF URL per camera (or NVR/VMS host, channel list, API endpoint)	The relay needs exactly one way in per camera	Yes
Credentials, and the policy for rotating them	Held server-side in the secret store; never in a browser	Yes
Network path: VLAN/subnet, firewall rule needed from the district edge	Analytics runs beside the cameras, not over the internet	Yes
Camera make, model, codec, resolution, frame rate	Tiering and capacity planning; read from the stream where not supplied	Preferred
Installation year, AMC or warranty expiry, firmware version	Ageing-infrastructure gap analysis and replacement planning; matching vendor security advisories	Preferred
Recorder type and retention days, storage location (local/cloud)	Retention reporting and evidence retrieval	Preferred
Which cameras face traffic at a readable angle	Sets `anpr_viable`, and therefore which tier the camera lands in	Preferred
Ownership: government or privately shared, and the sharing agreement	Access scoping and the audit trail	Preferred
Existing VMS and version, and whether RTSP re-streaming is permitted	Decides direct RTSP integration versus a vendor SDK adapter	If a VMS is involved

Anything in the "preferred" rows that is missing is inferred from the stream or left blank and reported in the gap analysis, so onboarding is never blocked on paperwork that can follow.

11. What the PoC must measure next

1. Streams per GPU-native node for tier A (replaces the 60-stream assumption), and the node's power draw (replaces 0.5 kW)
2. Keyframe-only decode density for tier B on the node (replaces 250). The profile is built; on the laptop it counted 40 streams at once on 2.46 CPU cores with the GPU at 13% (`bench/results/counts_capacity.md`), so the node's CPU, not its GPU, is the first thing to size. Its count accuracy against hand counts, and the load of both profiles on one node, are unmeasured too
3. Real plate-read accuracy on government footage by camera class, to set `anpr_viable`. First figure from the organisers' sandbox: 90 of 120 (75.0%) stored plates exact, 107 (89.2%) within one character, almost all on the 2 of 30 views where plates are readable, checked blind against their crops by a team member on the

control room's plate review page (`bench/results/sandbox_review.md`, 26 Sep). By Sentinel's confidence: 69 of 81 exact (85.2%) at 0.9 or above, 8 of 18 between 0.5 and 0.7, so high-confidence reads can pass and the rest go to a person; the page's bands set that floor and `anpr_viable` from real plates. An earlier check by our AI assistant, not blind, gave an optimistic 92 of 107. The plate-reading quality report (HLD 5.3) now tiers every camera from what it captures: on the sandbox, CAM06 and CAM07 are marginal (median plates 104 and 106 px across) and the other 28 counts-only, including the five the registry marks ANPR-viable (`bench/results/capture_quality.md`). Its grade describes the camera and does not predict a right read on real plates, so `anpr_viable` is set from the tier and blind reviews together. Still to measure: a second reviewer, the capture rate and the other camera classes
4. Vehicles per ANPR camera per day on real junctions (replaces 6,000)
5. Actual tier shares as departments onboard their fleets through the registry
6. Core ingest, matching and search at peak rates on server hardware, with replication and several API instances (556 sightings/s and 400 count rows/s on average, higher at peak). One laptop sustained 706 sightings/s against the 556/s average on 24 Sep (`bench/results/core_scale.md`); count rows were not included
7. Camera bitrates per district from the registry and the site survey (replaces the 4 and 0.8 Mbps assumptions in §4.1)
8. Restore time from the first restore drill (replaces the RTO target in §7.1)

Appendix A. Model inputs

Printed by `python bench/capacity.py` with the values used in this document.

Input	Value	Kind	Basis
Fleet and tiers			
`--cameras`	80,000	GIVEN	approximate statewide estate in the problem statement
`--departments`	26	GIVEN	departments running CCTV today, per the problem statement
`--districts`	33	ASSUMPTION	one edge site and one control room per district; Gujarat has 34 revenue districts since Vav-Tharad was carved out in 2025, so confirm against police districts and commissionerates
`--anpr-share`	0.1	ASSUMPTION	share of cameras positioned for plate reads (tier A); the registry's anpr_viable replaces it
`--event-share`	0.2	ASSUMPTION	share with low-rate event analytics, i.e. counts (tier B)
`--anpr-fps`	5	ASSUMPTION	tier A analysis rate, the worker's target; the laptop achieved 4.8 per stream at 1-2 streams (17 Sep)
`--event-fps`	1	ASSUMPTION	tier B analysis rate, keyframes only; the built profile counts at most 1 frame/s per camera (`COUNTS_FPS`), 0.78 median at 40 laptop streams (7 Oct)
Edge and core compute			
`--node-anpr-streams`	60	ASSUMPTION	1080p ANPR streams per 1-GPU edge node at 5 fps; the first thing the PoC measures (sentinel-bench)
`--node-event-streams`	250	ASSUMPTION	keyframe-only sub-stream decodes per node at ~1 fps; the laptop counted 40 at once on 2.46 CPU cores (bench/results/counts_capacity.md, 7 Oct), a direction, not a node figure
`--headroom`	0.25	ASSUMPTION	N+1 failover and growth
`--node-kw`	0.5	ASSUMPTION	average draw of a 1-GPU 2U edge server under load
`--match-ms`	1.14	MEASURED	median watchlist match per plate read not on the list, 100,000 entries, one connection (bench/results/matcher.md, 24 Sep)
Traffic and payloads			
`--vehicles-per-anpr-cam-day`	6,000	ASSUMPTION	vehicles read per ANPR camera per day; real junction counts replace it
`--sighting-row-bytes`	600	ASSUMPTION	Postgres row + indexes; the PoC's populated partition holds 503 B per row incl. indexes (4,205 rows, 24 Sep)
`--spool-row-bytes`	400	MEASURED	a sighting as the worker spools it (JSON): 399 B average over 4,205 PoC sightings re-serialised in the sink's format, 24 Sep
`--crop-kb`	8	ASSUMPTION	JPEG plate crop; the PoC's 4,187 crops average 6.2 KB (24 Sep)
`--count-window-s`	60	GIVEN	PoC COUNT_WINDOW_S: one vehicle/person count row per camera per window
`--count-row-bytes`	600	MEASURED	512 B busiest PoC count row + ~80 B of index
`--spool-count-bytes`	430	MEASURED	a count row as the worker spools it (JSON): largest of 368 PoC rows re-serialised (331 B average), 24 Sep

Input	Value	Kind	Basis
`--wal-factor`	2	ASSUMPTION	WAL bytes written per stored byte (indexes, full-page writes)
Retention and recovery			
`--hot-days`	90	ASSUMPTION	sightings and counts kept in Postgres
`--crop-days`	30	ASSUMPTION	plate crops kept at the edge
`--archive-years`	5	ASSUMPTION	warm Parquet archive of sightings and counts on the object store
`--pitr-days`	14	ASSUMPTION	point-in-time recovery window (WAL kept at DR)
`--backup-fulls`	2	ASSUMPTION	weekly full base backups kept at DR
`--spool-mb`	512	GIVEN	PoC SPOOL_MAX_MB: store-and-forward cap per worker
`--spool-days`	7	ASSUMPTION	backhaul outage an edge node rides out without dropping sightings
`--model4-days`	15	ASSUMPTION	central recording retention for the Model 4 comparison: the upper end of the 7-15 days departments keep today
Viewing and backbone			
`--state-rooms`	1	ASSUMPTION	state-level control rooms, on top of one per district
`--tiles-per-room`	16	ASSUMPTION	video-wall tiles per control room at peak
`--preview-mbps`	0.8	ASSUMPTION	640p H.264 preview (the camera's sub-stream); also tier B's analysis input
`--main-mbps`	4	ASSUMPTION	1080p main stream; the PoC caps its real-footage clips at 4 Mbps
`--fullres-share`	0.1	ASSUMPTION	tiles opened at full resolution
`--district-link-mbps`	50	ASSUMPTION	each backhaul link per district
`--links-per-district`	2	ASSUMPTION	two carriers, so one link can fail
Costs (₹ lakh)			
`--node-cost-lakh`	15	ASSUMPTION	1-GPU edge server incl. storage; GeM quote needed
`--gpu-card-lakh`	4	ASSUMPTION	the GPU's share of the node price
`--gpu-refresh-years`	5	ASSUMPTION	accelerator replacement cycle
`--amc-pct`	8	ASSUMPTION	annual maintenance, % of hardware capex, once warranty ends
`--warranty-years`	3	ASSUMPTION	OEM warranty bought with the hardware
`--tariff-rs-kwh`	8	ASSUMPTION	electricity, ₹ per kWh
`--pue`	1.6	ASSUMPTION	district server-room overhead for cooling
`--rack-lakh-yr`	6	ASSUMPTION	State Data Centre co-location per rack-year incl. power and cooling; may be nil for a Home Department tenant

Input	Value	Kind	Basis
`--district-link-lakh-yr`	3	ASSUMPTION	one district backhaul link per year
`--core-links`	3	ASSUMPTION	SDC uplink, DR uplink, SDC-DR replication link
`--core-link-lakh-yr`	20	ASSUMPTION	one core link per year
`--core-fte`	10	ASSUMPTION	platform team: lead 1, SRE/NOC on-call 4, backend 2, analytics 1, DBA 1, security 1
`--core-fte-lakh`	18	ASSUMPTION	loaded cost per FTE-year
`--districts-per-engineer`	3	ASSUMPTION	field engineers for edge hardware
`--field-fte-lakh`	8	ASSUMPTION	loaded cost per field-engineer-year
`--dept-integration-lakh`	10	ASSUMPTION	per department: ~2 person-months plus site visits
`--pilot-lakh`	30	ASSUMPTION	wave 0 field costs; team time is in opex
`--hardening-fte-years`	3	ASSUMPTION	PoC-to-production engineering, at the platform FTE rate
`--install-lakh`	0.5	ASSUMPTION	per edge node: racking, cabling, OS hardening, commissioning
`--district-net-lakh`	10	ASSUMPTION	per district: 10 GbE aggregation and fibre from camera VLANs to the edge rack; the site survey replaces it
`--seats-per-room`	2	ASSUMPTION	new operator workstations per control room
`--workstation-lakh`	1.5	ASSUMPTION	one operator workstation
`--training-lakh`	2	ASSUMPTION	per control room: operator and supervisor batches
`--audit-lakh`	25	ASSUMPTION	STQC / CERT-In empanelled security audit before go-live
`--years`	5	ASSUMPTION	total-cost-of-ownership horizon